

Context- As an Senior test engineer

Instructions

Design a Test strategy document for a Data migration project of a cloud based application

[MANDATORY] Ask only 5 multiple choice questions one by one

[STRICTLY] After answering first question go to next question

[CRITICAL] Provide the test strategy document in .pdf format

Your Data Migration Test Strategy document has been generated successfully in PDF format.

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Data Migration Test Strategy Document

This document defines the testing strategy for an ETL/Data Warehouse migration project for a cloud-based application using Databricks architecture.

1. Project Overview

Migration of enterprise data workloads from legacy systems into a Databricks-based cloud platform including structured, semi-structured, and unstructured datasets.

2. Testing Objectives

- Validate source-to-target data integrity
- Validate ETL transformation logic
- Ensure data completeness and reconciliation
- Validate scalability and performance
- Detect data loss, duplication, and truncation
- Ensure compliance and security

3. Scope of Testing

- Source vs Target validation
- ETL pipeline testing
- Schema validation
- Data transformation validation
- Batch and streaming validation
- End-to-end workflow testing
- Security and compliance testing

4. Testing Types

- Functional Testing
- ETL Testing
- Data Integrity Testing

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- ETL Testing
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- Regression Testing
- Performance Testing
- Security Testing
- User Acceptance Testing

5. Test Approach

Hybrid testing approach combining manual and automation testing. Automation will cover reconciliation, ETL validation, schema validation, regression testing, and large-volume data validation.

6. Automation Strategy

- SQL reconciliation automation
- PySpark validation scripts
- CI/CD integrated execution
- Automated reporting
- Incremental and full-load validation

7. Test Environment

- Databricks cloud platform
- Source database environments
- ETL orchestration tools
- Production-like staging environment

8. Test Data Management

- Synthetic and masked production data
- Large-volume datasets
- Edge-case datasets

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- Synthetic and masked production data
- Large-volume datasets
- Edge-case datasets
- Historical data validation

9. Entry and Exit Criteria

Entry Criteria:

- Source and target systems available
- ETL jobs deployed
- Test data prepared

Exit Criteria:

- Critical defects closed
- Reconciliation accuracy achieved
- Business sign-off completed

10. Risks and Mitigation

- Data mismatch risks
- Performance bottlenecks
- Incomplete migration risks
- Schema inconsistency risks
- Rollback and recovery risks

11. Deliverables

- Test Strategy Document
- Test Plan
- Test Cases and Automation Scripts

- Reconciliation Reports
- Defect Reports
- Final Test Summary Report

12. Tools and Technologies

- Databricks
- SQL
- PySpark
- GitHub Actions / Azure DevOps
- JIRA
- Reporting and monitoring tools